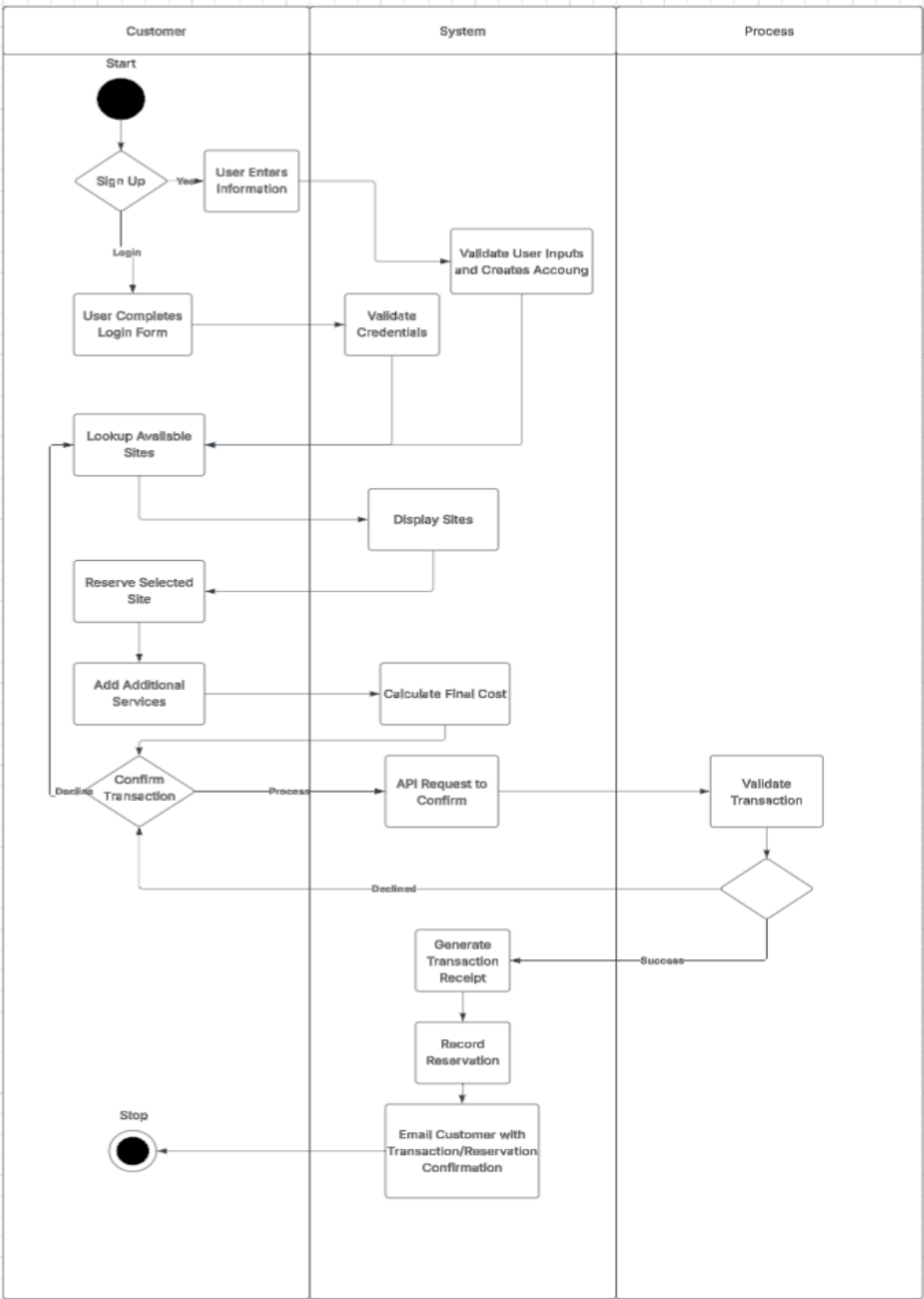
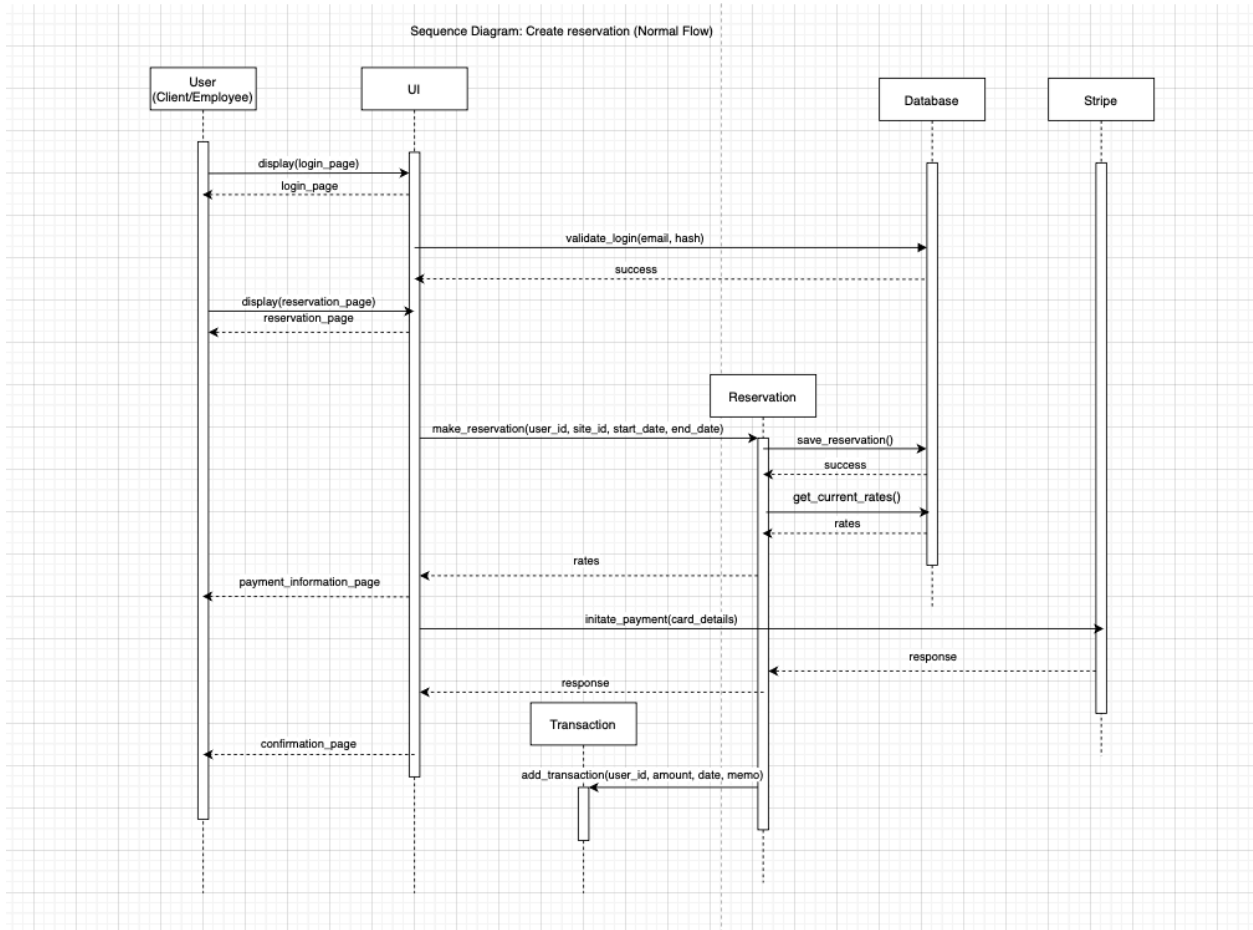


Updated Diagrams

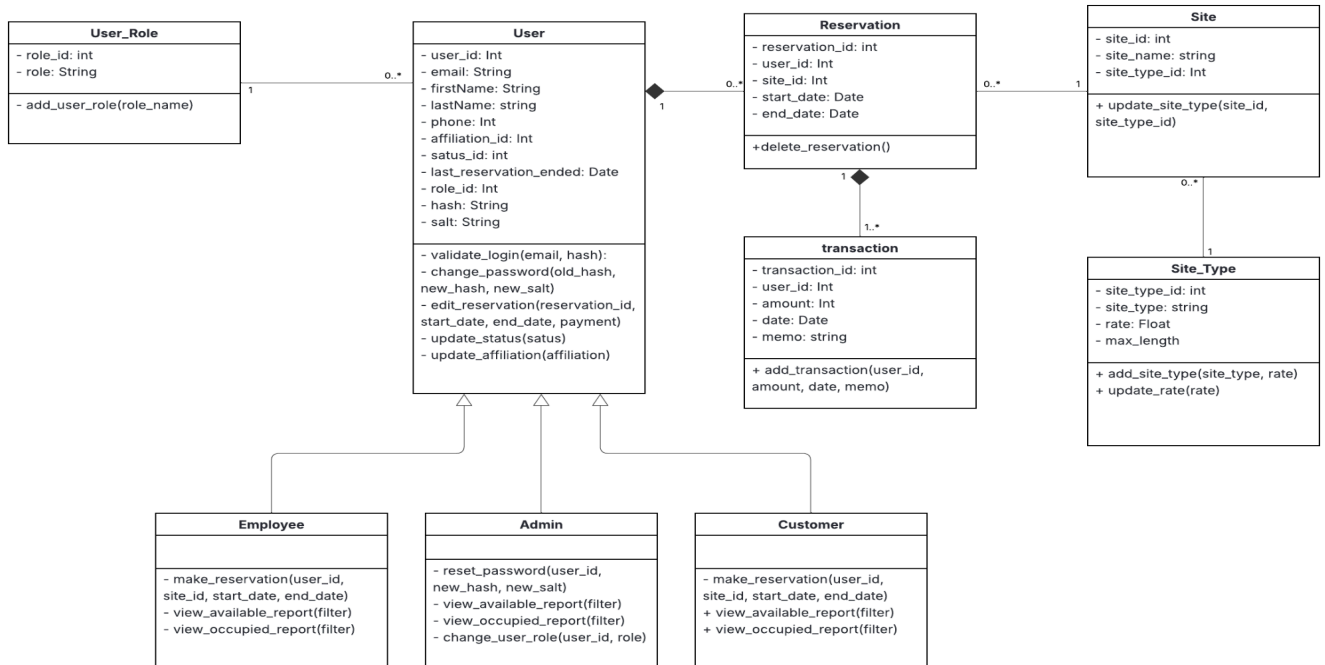
Activity Diagram:



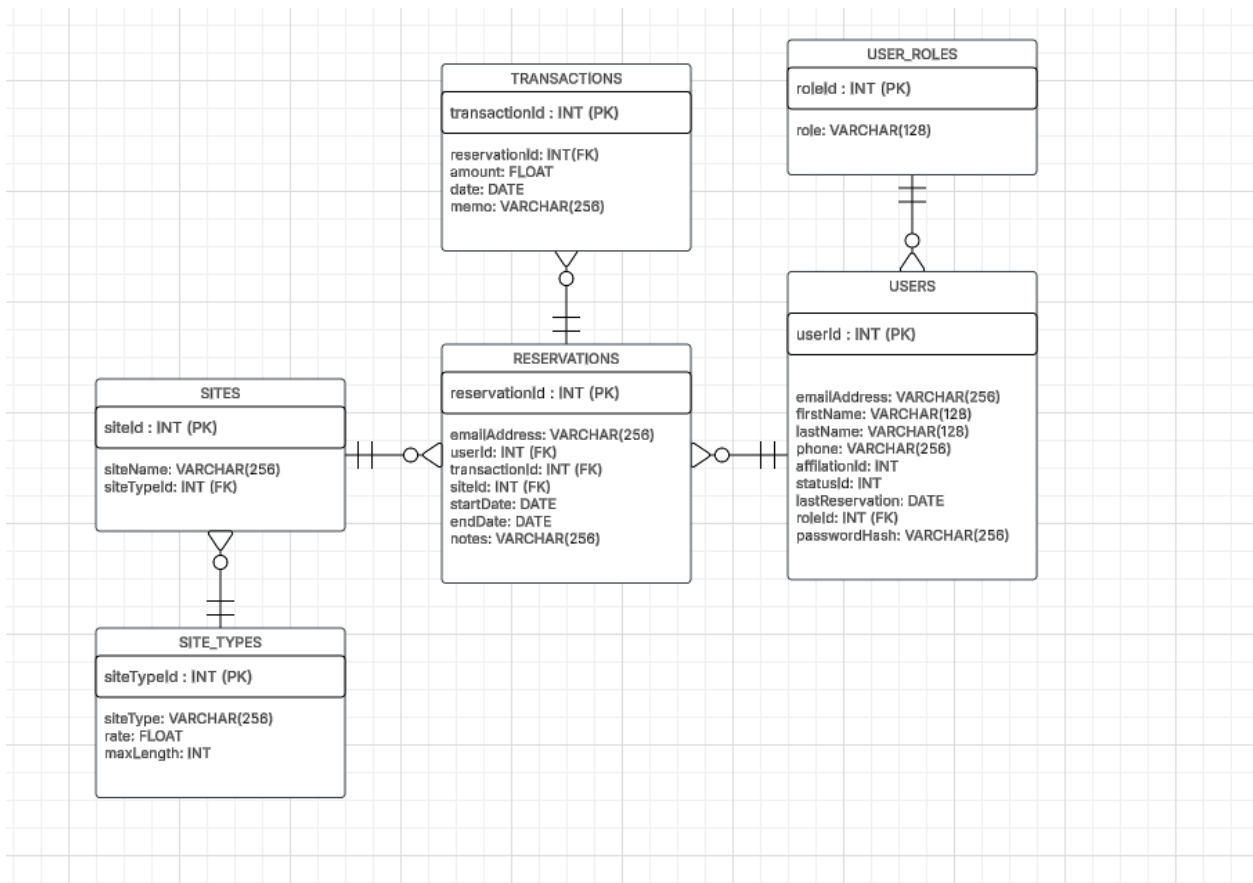
Sequence Diagram:



Class Diagram:

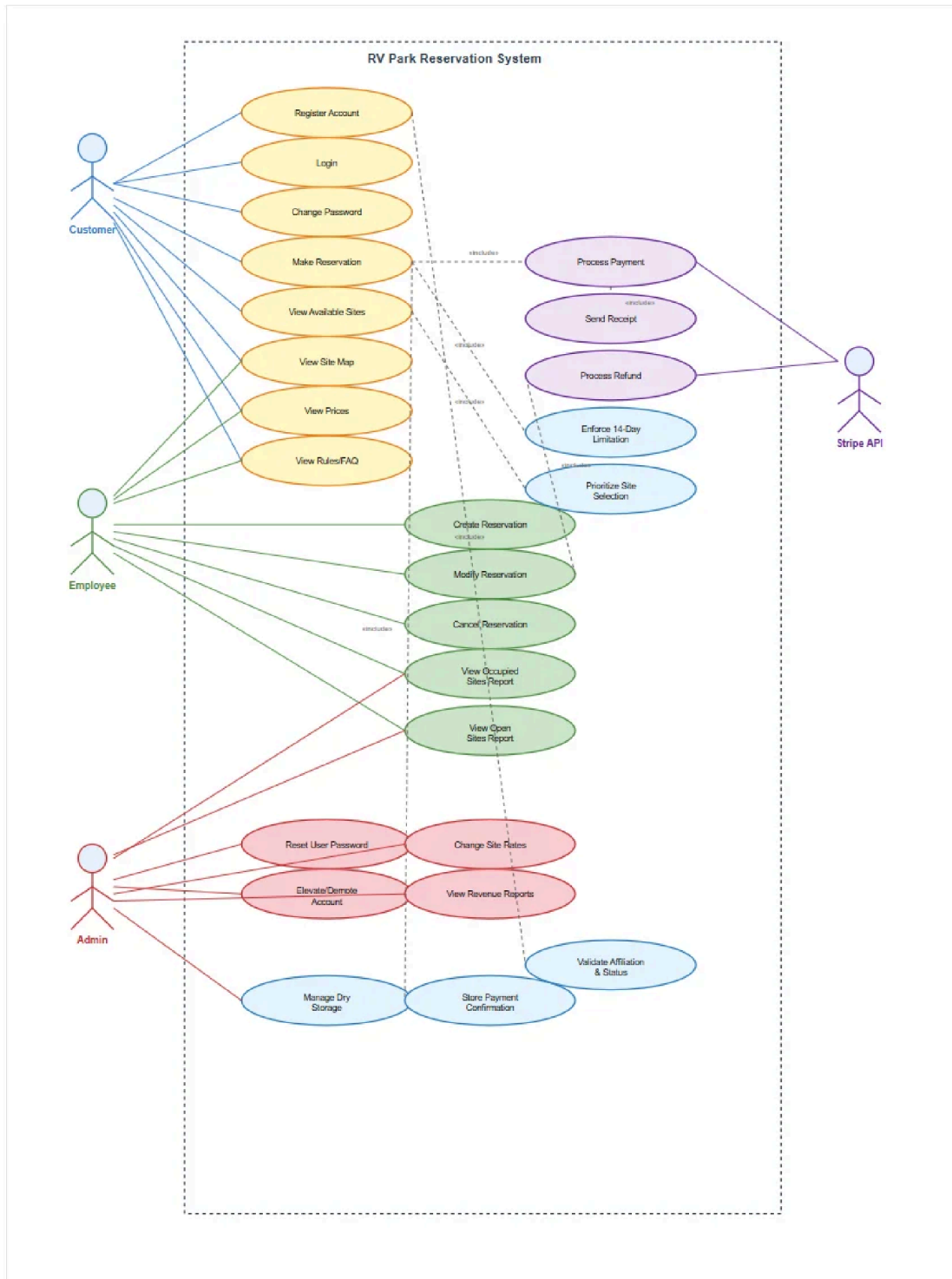


ERD:



Use Case Diagram:

RV Park Reservation System - Use Case Diagram



FUNCTIONAL REQUIREMENTS

F - FUNCTIONALITY

1. Account Management

1.1 Customer Account Features

- Customers shall be able to register their own account with the following required fields:
 - First Name
 - Last Name
 - Email Address
 - Phone Number
 - Password
- Customers shall be able to log in using their email and password.
- Customers shall be able to change their password by providing their existing password.

1.2 Employee Account Features

- Employees shall have assigned roles in the system.
- Employees shall be able to log in to access employee-specific features.

1.3 Admin Account Features

- Admins shall be able to reset user passwords to a default password.
- Admins shall be able to elevate user accounts to employee or admin roles.
- Admins shall be able to demote employee or admin accounts.

2. Reservation Management

2.1 Customer Reservation Features

- Customers shall be able to view available sites on a calendar interface.
- Customers shall be able to make reservations up to 6 months in advance only.
- Customers shall be able to select reservation dates (check-in and check-out).
- The system shall prioritize site selection to use the smallest available space that fits the customer's RV.
- Customers shall be able to view a site map showing all available spaces.
- Customers shall be able to view current pricing for different site categories.

- Customers shall be able to view rules and FAQ information.

2.2 Site Categorization

- The system shall support the following RV site categories:
 - Sites 1-14: 42'-43' long (max RV length 40')
 - Sites 17-31: 45'-46' long (max RV length 43')
 - Sites 32-45: 65' long
 - Tent/RV Sites: 1, 19, 21
 - Tent-Only Sites: 1 site
 - Sites 11 and 11B: Separate reservable sites
 - Sites 12 and 12B: Separate reservable sites
- Each site shall be assigned to a category (Dry Camp, Tent, Premium, Normal) that determines pricing.
- All sites within the same category shall have the same nightly rate.

2.3 Employee Reservation Management

- Employees shall be able to create new reservations for customers.
- Employees shall be able to modify existing reservations (including extending stays).
- Employees shall be able to cancel reservations.
- Employees shall be able to view all reservations in the system.
- Employees shall be able to process payments for reservations.
- Employees shall be able to view the site map.
- Employees shall be able to view current pricing.
- Employees shall be able to access rules and FAQ information.

3. Payment Processing

3.1 Payment Integration

- The system shall integrate with Stripe API for payment processing.
- The system shall pass the following information to Stripe:
 - Customer ID
 - Payment Date
 - Payment Amount
- The system shall store the following payment information:
 - Payment Confirmation ID (from Stripe)
 - Payment Date
 - Payment Type
 - Payment Amount

- Payment Status
- The system shall process refunds through Stripe.

3.2 Receipt Management

- The system shall automatically send a receipt to customers after successful payment.
- Employees shall be able to manually send receipts to customers.

4. Reporting

4.1 Admin Reporting Features

- Admins shall be able to view an "Open Sites Report" containing:
 - Available/Reserved status for each site
 - Number of nights available
 - Hook-up type
 - Rate per night for each date
- Admins shall be able to view an "Occupied Sites Report" containing:
 - Date
 - Customer Last Name
 - Site Number
 - Number of Nights
 - Reservation Notes (if any)
- Admins shall be able to filter all reports by date range.
- Admins shall be able to view revenue reports.

4.2 Employee Reporting Features

- Employees shall be able to view the same reports available to admins.

5. Site Rate Management

5.1 Pricing Administration

- Admins shall be able to change the nightly rate for each site category.
- Rate changes shall apply to future reservations only.

6. Dry Storage Management

6.1 Storage Tracking

- The system shall support dry storage (vehicle/RV parking) separate from reservations.
- The system shall store the following information for dry storage:
 - Customer ID
 - Beginning Date
 - End Date
 - Amount Paid
- Admins shall be able to manage dry storage records.

7. Business Rules

7.1 Reservation Constraints

- The system shall enforce the 14-day limitation only during peak season (April-October).
 - Reservations that begin or end outside peak season shall not be subject to the 14-day limitation.
 - The system shall validate affiliation and status for all users during registration.
 - The system shall prioritize site assignments to minimize wasted space (smallest fitting site first).
-

NON-FUNCTIONAL REQUIREMENTS

U - USABILITY

8. User Interface

8.1 Accessibility

- The system shall be fully functional on mobile devices (responsive design).
- The user interface shall be intuitive and require minimal training for customers.
- Navigation between pages shall be clear and consistent across all user roles.

8.2 User Experience

- The system shall provide clear error messages when validation fails.
- The system shall provide confirmation messages for all successful actions (reservations, payments, account updates).
- The site map shall be easy to read and understand.

- The reservation process shall be completable in 5 minutes or less for experienced users.

8.3 Help and Documentation

- Rules and FAQ information shall be easily accessible from all user interfaces.
- The system shall provide contextual help where complex operations occur.

R - RELIABILITY

9. System Availability

9.1 Uptime

- The system shall maintain 24-hour availability with 99.5% uptime.
- Planned maintenance downtime shall be minimal and scheduled during low-usage periods.

9.2 Data Integrity

- All reservation data shall be stored reliably without loss.
- Payment confirmations shall be stored permanently and remain accessible.
- The system shall prevent double-booking of sites.

9.3 Error Handling

- The system shall handle payment processing failures gracefully and inform users.
- The system shall log all errors for administrative review.
- The system shall recover from temporary Stripe API failures without data loss.

9.4 Backup and Recovery

- The system shall perform daily automated backups of all data.
- The system shall be recoverable within 4 hours in case of catastrophic failure.

P - PERFORMANCE

10. Speed and Responsiveness

10.1 Page Load Times

- All pages shall load within 3 seconds under normal network conditions.

- The site availability calendar shall load within 2 seconds.

10.2 Transaction Processing

- Payment processing shall complete within 10 seconds (excluding Stripe processing time).
- Reservation creation shall complete within 5 seconds.

10.3 Scalability

- The system shall support at least 100 concurrent users without performance degradation.
- The system shall handle peak booking periods (beginning of 6-month reservation window) efficiently.

10.4 Database Performance

- Report generation shall complete within 10 seconds for date ranges up to 1 year.
- Search and filter operations shall return results within 2 seconds.

S - SUPPORTABILITY

11. Maintainability

11.1 Code Quality

- The system shall be built with modular, well-documented code.
- The system shall follow established coding standards and best practices.

11.2 System Updates

- The system shall support updates without requiring extended downtime.
- Database schema changes shall be managed through migration scripts.

11.3 Monitoring and Logging

- The system shall log all user actions for audit purposes.
- The system shall log all payment transactions.

- System administrators shall have access to monitoring dashboards showing system health.

12. Testability

12.1 Testing Support

- The system shall support automated testing for critical functions.
- The system shall include a test environment separate from production.
- All business rules shall be testable independently.

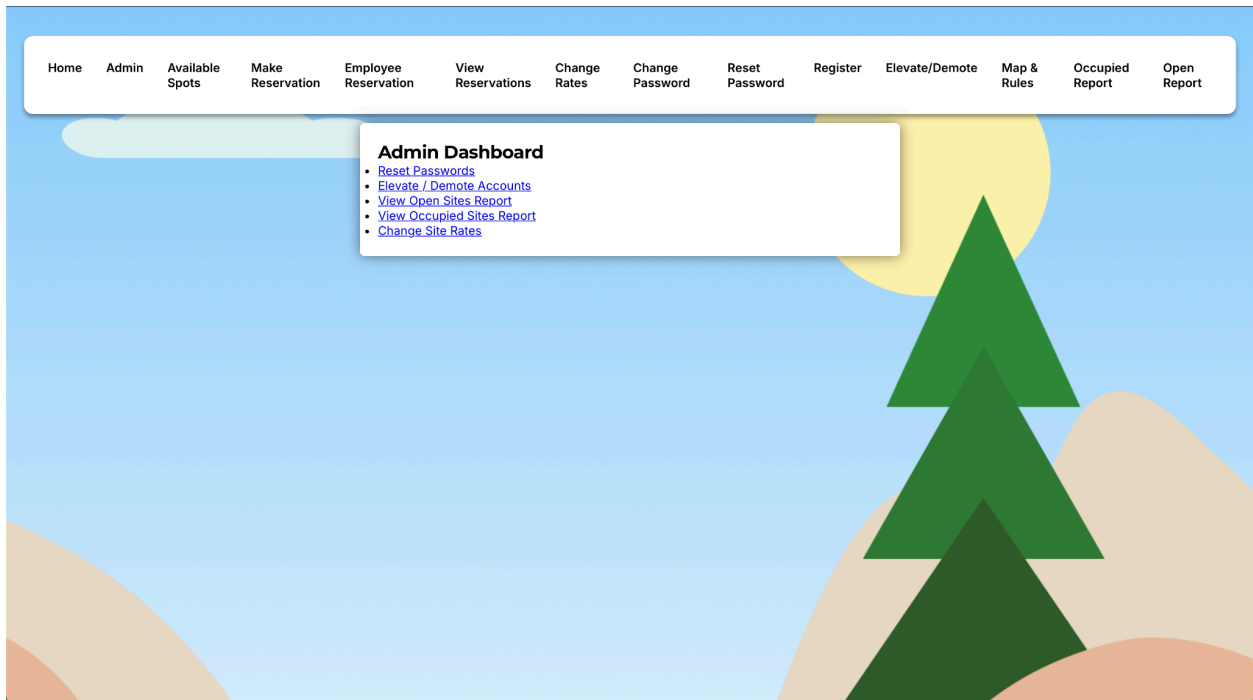
13. Configurability

13.1 System Configuration

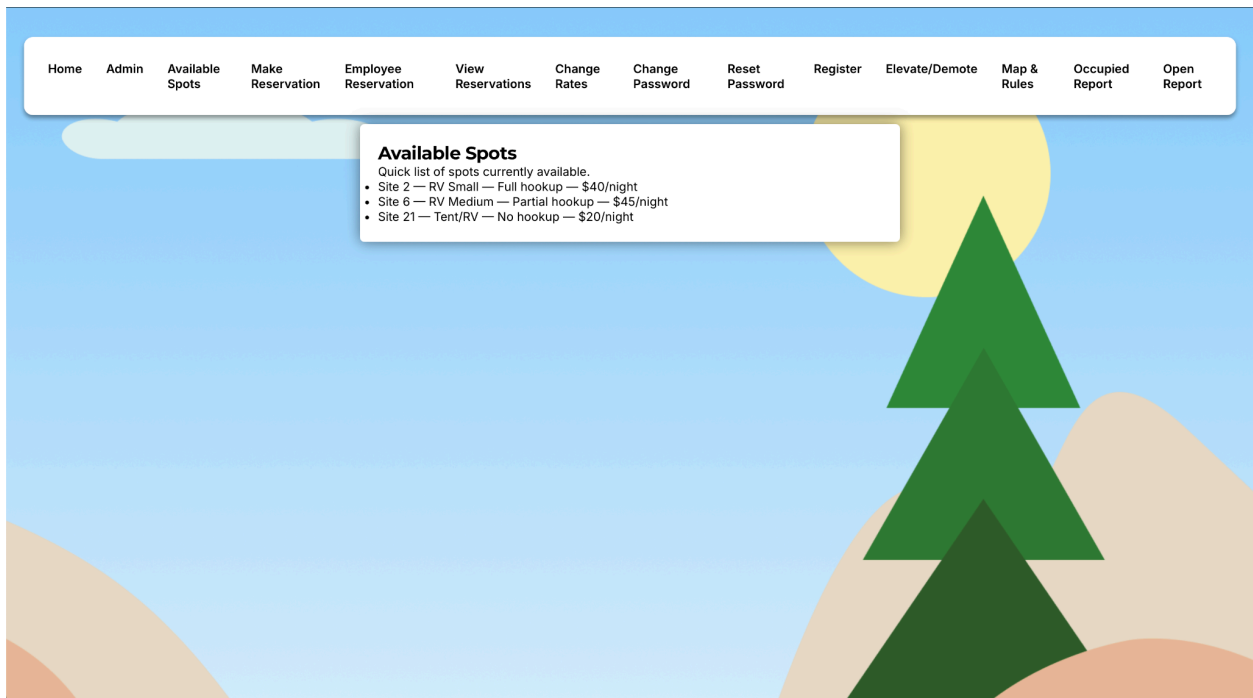
- Site categories and pricing shall be configurable without code changes.
- The 6-month advance reservation limit shall be configurable.
- Peak season dates shall be configurable by administrators.

Front End Screens:

Admin Dashboard:



Available Spots:



Make Reservation:

HomeAdminAvailable SpotsMake ReservationEmployee ReservationView ReservationsChange RatesChange PasswordReset PasswordRegisterElevate/DemoteMap & RulesOccupied ReportOpen Report

Make a Reservation

Name

RV size (ft)

Site size (ft)

Start date

11/08/2025

End date

11/08/2025

Time

12:30 PM

Submit Reservation

localhost:8773/make-reservation

View Reservations:

HomeAdminAvailable SpotsMake ReservationEmployee ReservationView ReservationsChange RatesChange PasswordReset PasswordRegisterElevate/DemoteMap & RulesOccupied ReportOpen Report

View / Edit Reservations

List of current reservations with basic edit links.

Reservation ID	Name	Site	Start	End	Actions
123	Jane Doe	12	2025-11-01	2025-11-03	Edit

Change Rates:

HomeAdminAvailable SpotsMake ReservationEmployee ReservationView ReservationsChange RatesChange PasswordReset PasswordRegisterElevate/DemoteMap & RulesOccupied ReportOpen Report

Change Rates

Update nightly rates for each site type.

RV Small (1-14):

RV Medium (17-31):

RV Large (32-45):

Save Rates

Change Password:

HomeAdminAvailable SpotsMake ReservationEmployee ReservationView ReservationsChange RatesChange PasswordReset PasswordRegisterElevate/DemoteMap & RulesOccupied ReportOpen Report

Change Your Password

Current Password:

New Password:

Confirm New Password:

Change Password

Reset Password:

HomeAdminAvailable SpotsMake ReservationEmployee ReservationView ReservationsChange RatesChange PasswordReset PasswordRegisterElevate/DemoteMap & RulesOccupied ReportOpen Report

Reset Password

Account Email or Username:

New Password:

Confirm New Password:

Reset Password

Register:

HomeAdminAvailable SpotsMake ReservationEmployee ReservationView ReservationsChange RatesChange PasswordReset PasswordRegisterElevate/DemoteMap & RulesOccupied ReportOpen Report

Register Page

Username:

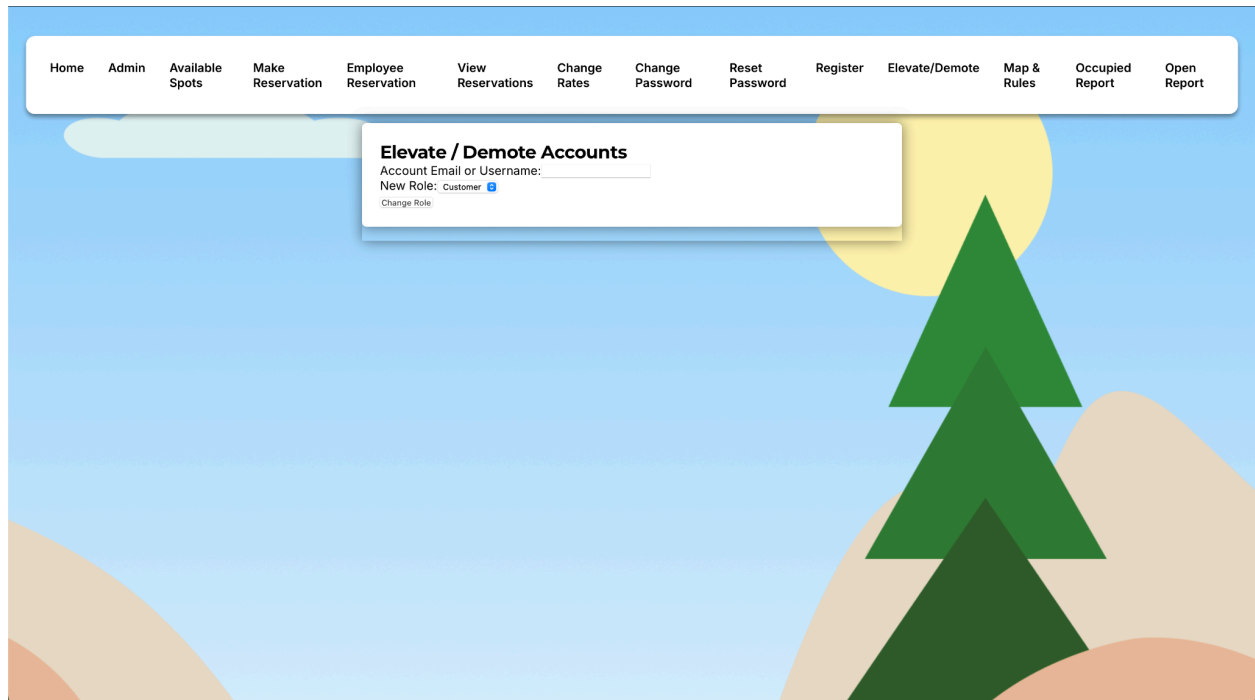
Password:

Confirm Password:

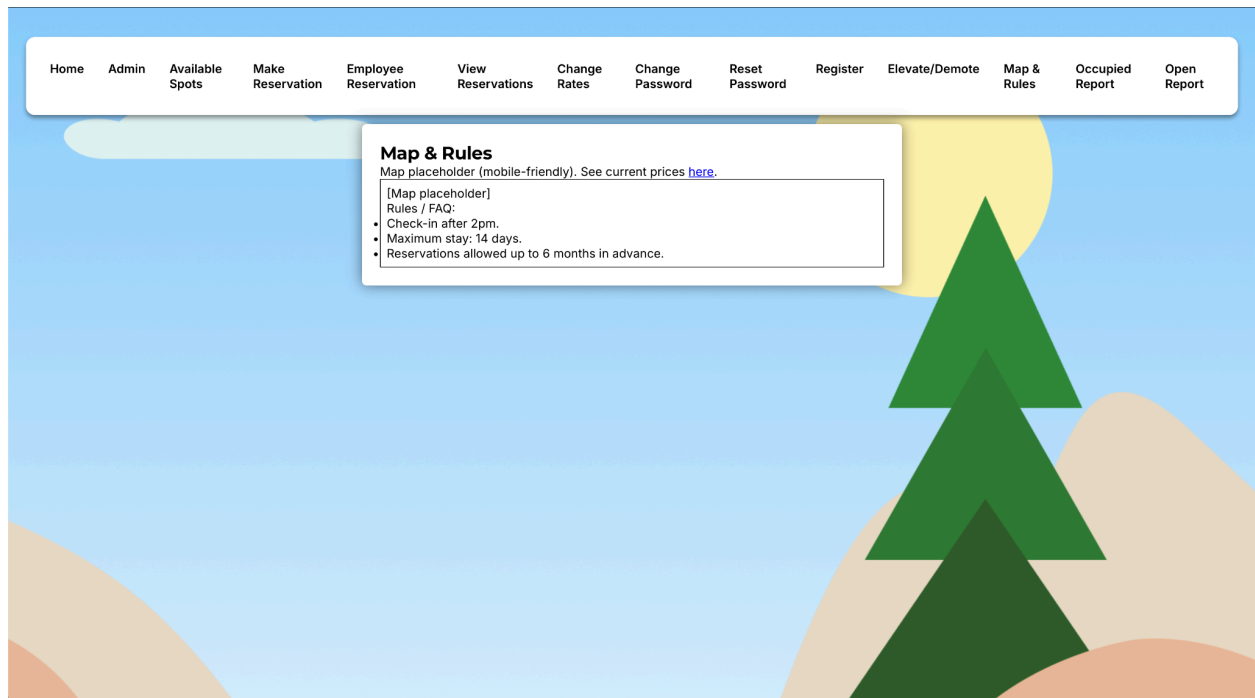
Rank:

Register

Elevate/Demote:



Map & Rules:



Occupied Report:

[Home](#) [Admin](#) [Available Spots](#) [Make Reservation](#) [Employee Reservation](#) [View Reservations](#) [Change Rates](#) [Change Password](#) [Reset Password](#) [Register](#) [Elevate/Demote](#) [Map & Rules](#) [Occupied Report](#) [Open Report](#)

Occupied Sites Report

Shows date, last name, site, nights, and reservation notes.

Date	Last Name	Site	Nights	Notes
2025-11-01	Smith	12	2	Arriving late

Open Report:

[Home](#) [Admin](#) [Available Spots](#) [Make Reservation](#) [Employee Reservation](#) [View Reservations](#) [Change Rates](#) [Change Password](#) [Reset Password](#) [Register](#) [Elevate/Demote](#) [Map & Rules](#) [Occupied Report](#) [Open Report](#)

Open Sites Report

Shows which sites are available / reserved, nights available, hook-up, and rate per night for each date.

Date	Site	Available Nights	Hook-up	Rate
2025-11-05	Site 53	Full		\$45